DEBANSHU SARKAR

CPU Architecture | Embedded Systems | Instrumentation | Firmware | Systems Software

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SUMMARY

Electronics and Instrumentation Engineering student (B.Tech, MAKAUT, 2027) ranked 3rd in department with hands-on experience building and debugging software systems across multiple hardware platforms. Proficient in C, C++, Java, Python and JavaScript with strong OOP design skills. Experienced with multi-platform systems design, testing, and debugging on Linux and microcontroller environments. Built and shipped published software tools used by hundreds of developers. Comfortable working across the full development cycle from design and coding through test execution and defect resolution. Also have knowledge about CPU architectures ranging from CISC, RISC, 8085, 8086, ATMega328 and ARM processors.

EDUCATION

B.Tech, Electronics and Instrumentation Engineering 2023 – 2027

Techno Main Salt Lake, MAKAUT, Kolkata | Ranked 3rd in Department | No Academic Backlogs

Relevant Coursework: Embedded Systems, Microprocessors & Microcontrollers, Digital Signal Processing, IoT & Wireless Sensor Networks, Control Systems, Analog & Digital Electronics, Network Theory, Sensors & Transducers, Database Management Systems, Data Structures & Algorithms, Object-Oriented Programming

TECHNICAL SKILLS

Languages: C, C++, Embedded C, Java, Python, JavaScript, Assembly (8085/8086), Verilog

OOP & Design: Object-Oriented Design, Design Patterns, Modular Architecture, Interface Abstraction, Multithreading

Systems & Platforms: Linux (Raspberry Pi OS), Embedded Linux, FreeRTOS / Zephyr (learning), ESP32, Arduino, Raspberry Pi 4B

Protocols & Interfaces: UART, I2C, SPI, GPIO, PWM, ADC, MQTT, WebSocket, MAVLink, CAN (learning)

Software Design Tools: Git/GitHub, VS Code, EasyEDA, MATLAB, systemd daemon management

Testing & Debugging: Logic analyser, oscilloscope, unit testing, GitHub Actions CI/CD, multithreading and race condition debugging

Databases & Cloud: SQLite, MySQL, Firebase, ThingSpeak IoT Cloud

AI & Vision: OpenCV, YOLOv8, MobileFaceNet, Vosk STT, Whisper, Llama 3.2 on-device LLM

SHIPPED SOFTWARE

Code Janitor | Node.js, JavaScript, VS Code Extension API, Arduino IDE 2.x | github.com/Debanshu2005/code-janitor2024 – Present

- Published VS Code extension with 200+ installs and 1,500+ total acquisitions, 5-star rated on VS Code Marketplace; designed, coded and maintained across versions v0.9.7 to v1.12.0 with 236 commits.
- Implemented core features: AI-powered code formatting, syntax validation, codebase graph visualisation, live preview, and an Arduino IDE-specific AI agent with structured FILE/PATCH/CMD actions.
- Integrated multiple AI providers (Ollama, Groq, Anthropic, NVIDIA NIM) with self-healing performance monitoring and persistent workspace memory; debugged and resolved provider compatibility issues across platform types.

IntelScan | Python, PyPI, GitHub Actions CI/CD | github.com/Debanshu2005/IntelScan 2025

- Published Python CLI tool on PyPI (pip install intelscan) that auto-generates workspace manifests and AI agent context files; includes full test suite, watch mode, and VS Code task integration.
- Set up GitHub Actions CI/CD pipeline with PyPI Trusted Publishing for automated build, test, and release workflows.

PROJECTS

DronePy — Onboard Drone AI Runtime | Python, MAVLink, LoRA Fine-tuning, GGUF | github.com/Debanshu2005/DronePy 2025 – Present

- Designed a modular Python runtime for autonomous drones with hardware discovery, flight controller probing (MAVLink/DroneCAN), and normalized capability profiling using OOP interfaces and abstract base classes.

- Built a learned mission planner pipeline with a structured feature payload; includes LoRA fine-tuning scripts, JSONL planner dataset schema, planner evaluation framework, and GGUF export for on-device inference.
- Implemented a deterministic safety boundary layer (battery guard, link health checks, capability gating) and a dry-run-friendly action executor; designed for Raspberry Pi deployment with CSI camera and GPIO integration.

Autonomous Quadcopter | APM 2.8, Raspberry Pi 4B, Python, MAVLink, DroneKit, ESP32 |
github.com/debadityap/Drone_software 2024 – Present

- Designed and implemented a Python mission control layer over ArduPilot via MAVLink covering autonomous waypoint navigation, pre-flight calibration, compass anchoring, and arming safeguards.
- Implemented voltage-gated arming logic, IMU anomaly-triggered soft landing, and LoRa SX127x wireless telemetry with an ESP32 ground station over UART.

Buddy — Mobile AI Robot | Raspberry Pi 4B, Arduino, YOLOv8, Llama 3.2, OpenCV, UART, systemd |
github.com/Debanshu2005/Pi_Buddy 2024 – Present

- Built a modular software system on Linux: on-device LLM (Llama 3.2 3B), Whisper STT, YOLOv8 object detection and MobileFaceNet face recognition, each deployed as a systemd daemon for process isolation and reliability.
- Separated perception, reasoning and actuation into independent modules in a ROS-compatible layout; Arduino-controlled 4WD platform driven over USB-serial UART.

ACHIEVEMENTS

- Brainwave 2.0 Hackathon Finalist, Delhi Technological University (DTU)
- Smart India Hackathon (SIH) 2024 and 2025 Participant; National Robotics Hackathon Participant
- Ranked 3rd in Department of Electronics and Instrumentation Engineering, 2nd Year; no academic backlogs
- 15+ public GitHub repositories spanning systems software, embedded systems, robotics, computer vision and published developer tools — github.com/Debanshu2005